

Course name: Technology and the Making of Modern India

Course code: HM 495

Course type: Open elective

Instructor: Purbasha Das

Credit: 3-0-0-3

Semester: Winter Semester 2026

Course Outline

This course seeks to explore the various connotations of technology and comprehend how it came to be seen as a defining feature of modernity. Technologies and scientific thinking were indispensable for the building of the modern Indian nation state. The objective is to observe how the nationalist and the colonial understanding of science and technology relate to each other, as well as how the governing classes' perspective changes over time. The Global South and the Global North now engage in an interchange of labour and knowledge instead of the former metropole and colony. The readings will allow the students to reflect on the Eurocentric notion that scientific and technological knowledge always flows from the Global North to the Global South. The goal is to understand the uniqueness of the Indian experience of science and technology while at the time understanding whether this experience can be called a part of the broader global realm of science and technology. The readings explore the interconnection between science and technology and larger processes of nationalism, colonialism and industrialization. They also point to other intersections with environment, labour and health. The key takeaway of the course is that technology is imbued with meaning and technological change impacts and is shaped by all levels of society.

Aims and Objectives

- The course will enable students to think critically about technology beyond its technical applications.
- The students will be introduced to a historical understanding of science and technology in the post-colonial period.
- Class discussions will unpack key concepts like nation, modernity, labour and technological progress.
- The students will be able to understand the favourable and unfavourable impact of technology on the economy, environment and other facets of society. The course will aid them in thinking about developing technology that is beneficial for different sections of society.
- The writing assignments and project will improve their written and oral communication skills.

Reading List (Selected) *

- 1) Introduction: Science as a Reason of State” in Ashish Nandy ed. *Science, Hegemony and Violence: A Requiem for Modernity* (New Delhi: Oxford University Press, 1988).
- 2) Arun Mohan Sukumar, *Midnight's Machines: A Political History of Technology in India* (Gurgaon: Viking by Penguin Random House India, 2019).
- 3) Sumit Sarkar, *Modern Times: India 1880s- 1950s: Environment, Economy and Culture* (Ranikhet: Permanent Black, 2014)
- 4) David Arnold, “Nehruvian Science and Postcolonial India,” *Isis*, Vol. 104, No. 2 (June 2013), pp. 360-370.
- 5) S. Irfan Habib, “Legacy of the Freedom Struggle: Nehru's Scientific and Cultural Vision,” *Social Scientist*, Vol. 44, No. 3/4 (March–April 2016): 29-40.
- 6) Asif A. Siddiqi, “Whose India? SITE and the origins of satellite television in India,” *History and Technology*, Vol.36, No.3-4, 2020, pp. 452-474.
- 7) Jahnvi Phalke, *Atomic State: Big Science in Twentieth-Century India* (Ranikhet: Permanent Black, 2013).
- 8) Nikhil Menon, “ ‘Fancy Calculating Machine’: Computers and Planning in IndependentIndia,” *Modern Asian Studies*, Vol.52, Issue 2 (March 2018), pp. 421-457.
- 9) Elizabeth Chatterjee, “The Poor Woman’s Energy: Low-Modernist Solar Technologies and International Development, 1878-1966,” *Journal of Global History*, Vol. 18, No.3, 2023, pp. 439-460.
- 10) Govindan Parayil, “The Green Revolution in India: A Case Study of Technological Change,” *Technology and Culture*, Vol. 33, No. 4, 1992, pp. 737–56.
- 11) Madhumita Saha, “The State, Scientists, and Staple Crops: Agricultural ‘Modernization’ in Pre–Green Revolution India.” *Agricultural History*, Vol. 87, No. 2, 2013, pp. 201–23.
- 12) Rohan D’Souza, “Supply-Side Hydrology in India: The Last Gasp,” *Economic and Political Weekly*, Vol. 38, No. 36, Sep. 6-12, 2003, pp. 3785-3790.
- 13) “A Dam in the Khasi Hills: The Umiām Hydroelectric Project 1960-1965” in William A.T. Logan ed., *A Technological History of Cold-War India, 1947-1969: Autarky and Foreign Aid* (Cham: Springer, 2022).

- 14) Amit Prasad, "Burdens of the Scientific Revolution: Euro/West-Centrism, Black Boxed Machines, and the (Post) Colonial Present," *Technology and Culture*, Vol. 60, No. 4, October 2019, pp.1059-1082.
- 15) Nasir Tyabji, "Negotiating Nonalignment: Dilemmas Attendant on Initiating Pharmaceutical Production in India," *Technology and Culture*, Vol.53, No.1, January 2012, pp. 37-60.
- 16) Madhav Gadgil and Ramchandra Guha, "Ecological Conflicts and the Environmental Movement in India" in Mahesh Rangarajan ed., *Environmental Issues in India: A Reader* (Delhi: Pearson Longman, 2007).
- 17) Madhav Gadgil, "Ecology: Modern Science and Traditional Wisdom", in Michael Tobias, Teun Timmers and Gill Wright ed., *A Parliament of Science: Science for the 21st Century* (Albany: State University of New York Press, 2003).
- 18) Ross Bassett, *The Technological Indian* (Cambridge, Massachusetts: Harvard University Press, 2016).
- 19) Aparajith Ramnath, *The Birth of an Indian Profession: Engineers, Industry and the State*. (New Delhi: Oxford University Press, 2017).

*The list includes readings which will be discussed in class.

Grading Policy

- Two in-semester exams (30% +30%)
- End semester project (40 %)

Course Outcomes:

Upon passing the courses, students should have a:

- Basic critical and analytical understanding of how technology is shaped by society and also how technology impacts society. [PO1, PO7, PO12]
- How the progress of technology is shaped by political, social, environmental factors. This can in turn be used to understand the ethics of technological production. [PO6, PO7, PO8].

POs-COs Matrix:

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
X					X	X	X				X

Course Outcomes and Competencies

Sr. No	Course Outcome	PO/PSO	CPD	KD
1	CO1: Basic critical and analytical understanding of how technology is shaped by society, polity and the economy.	PO1, PO7, PO12/PSO1	U	F,C
2	CO2: To understand the diversity of the impact of technology which is not limited to the technological sphere. The impact is observed in the everyday life of people in modern society.	PO6, PO7, PO8/PSO1, PSO2	U, Ap, An	F,C,P

CL: Cognitive Process Domain [R: Remember; U: Understand; Ap: Apply; An: Analyze; E: Evaluate; C: Create]

KC: Knowledge Domain [F: Factual; C: Conceptual; P: Procedural; M: Metacognitive]

Focus Area	Yes/No	Details
Focus on Employability	No	-
Focus on Skill development	Yes	The students develop necessary skills to develop knowledge in culture, society and identity.
Focus on entrepreneurship	No	-